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A DYNAMIC LOOK AT A CLASS OF SKEW DISTRIBUTIONS. A MODEL WITH SCIENTOMETRIC APPLICATIONS

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A theoretical model of repetitive events is presented and applied to the scientific publication process. Based on three simple postulates, a relation between population growth and distribution of authors by publication productivity in a scientific community is established. Predictions of the model are supported by empirical evidences.

Introduction

Habent sua fata distributiones. Irwin[1] was lead to study systematically inverse factorial series as frequency distributions in his attempt for "finding adequate theoretical forms to describe biological distributions with very long tails" and he suggested the name *Waring distribution* for the most suitable one. In order to illustrate the results, empirical data were needed. However, as *Irwin* established: "Good illustrative data are not esay to come by, when one wants them. . . I found a few, but the observed frequencies in the tails had always been grouped and the individual values were not presented. They were thus useless for my purpose. So the only illustrative examples I had were Dr. Kendall's from the bibliographic field." Later on, *Irwin*'s attention turned back to biological problems and accident theory[2,3], moreover, Waring distribution has found interesting applications in quantitative linguistics.[4] Nevertheless, it appears that the story of Waring distribution is inseparably connected with its bibliometric applications from the very beginnings up to the recent days.[5,6]

Interestingly enough, neither the above nor other authors did realize that the Waring distribution had already been proposed in the classic paper of *Simon*[7] as a possible generalization of the Yule distribution (cf. Eqs. (3.7) and (3.8) of the original paper). Although a vast literature having its origin in *Simon*'s paper has been published, a lot of them with definite bibliometric concern,[8–12] this generalization has apparently been overlooked all the time. *Simon*'s derivation (or rather, heuristic approach) of the Waring distribution is worth mentioning not only for its priority but also because the birth process he used in his approach seems to provide a much deeper insight into the "mechanism" of

processes leading to such distribution than does the compound distribution model of *Irwin* or *Tague*'s urn scheme. Considering this "mechanism", closely related to *Simon*'s scheme is that of *Frank*[13,14] leading to another limiting case of the Waring distribution: the geometric distribution. Several generation schemes of the Waring distribution, including those mentioned above, have been reviewed recently by *Xekalaki*.[15]

In this paper a full dynamic description of a model leading to Waring and related distributions is presented. This model can be regarded as a direct generalization of the schemes of *Simon* and *Frank* from "steady-state" (stationary) to "steady-change" (quasi-stationary) conditions. In more practical terms, we are interested not only in the time-independent internal structure of a given population but also in the temporal evolution of this structure, moreover, in the relation between gross changes in the population and the changes in its internal structure. The model, like its predecessors, was found to be applicable to a wide variety of fields. In this paper the power of the model is illustrated on the example of the classical scientometric problem of frequency distribution of scientific productivity.[16] Several good bibliographies and reviews are available on the topic,[17–21] so there is no use in giving a necessarily superficial survey. We only reiterate here the basic problem: given a population of scientists, we would like to know the distribution of persons authoring 0, 1, 2, . . . publications within a given period.

1. Theory

1.1. Outlines of the model

Consider an infinite array of cells (units, boxes, etc.) indexed in succession by the non-negative integers. Suppose that a (finite) amount, x, of a certain substance is distributed among the cells; let x_i denote the content of the i-th cell ($i \in N_0$). Obviously,

$$x = \sum_{i=0}^{\infty} x_i. \tag{1}$$

The fractions

$$y_i = x_i/x \quad (i \in N_0) \tag{2}$$

may as well be regarded as probability values of the distribution of a discrete random variable $\xi \in N_0$:

$$y_j = P(\xi = i) \quad (i \in N_0). \tag{3}$$

It will be required throughout this paper for this distribution to have a finite expected value:

$$E(\xi) = \sum_{i=0}^{\infty} iy_i. \tag{4}$$

The content of the cells may change due to the following processes (see Fig. 1):

(1) substance may enter the system from the external environment through the 0-th cell at a rate s;

(2) substance may be transferred unidirectionally from the i-th cell into the (i + 1)-th one at a rate f_i ($i \in \mathbf{N}_0$);

(3) substance may leak out from the i-th cell into the external environment at a rate g_i ($i \in \mathbf{N}_0$).

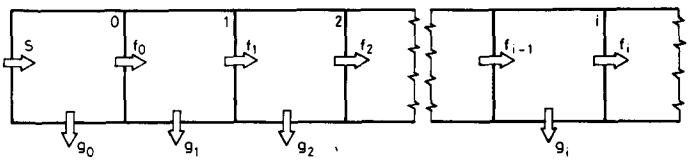


Fig. 1. Scheme of substance flow

The general equations of change of such a system will thus be

$$\begin{aligned} \dot{x}_0 &= s - f_0 \\ \dot{x}_i &= f_{i-1} - f_i - g_i \quad (i \in \mathbf{N}_0) \end{aligned} \tag{5}$$

(dot denotes time derivative).

Let us suppose now the following particular forms of s, f_i, and g_i:

$$s = \sigma x, \tag{6}$$

$$f_i = (\alpha + \beta i) x_i \quad (i \in \mathbf{N}_0), \tag{7}$$

$$g_i = \gamma x_i \quad (i \in \mathbf{N}_0), \tag{8}$$

where σ and α are positive, β and γ are non-negative real constants. In simple terms, Eq. (6) states that the system has a kind of "self-reproducing" property: the rate of external source is proportional to the actual total content of the system; Eq. (7) formalizes the so-called "cumulative advantage" property: the higher cell index is reached, the more facile the further transfer becomes; Eq. (8) asserts that "leakage" is uniform over the cells: its rate is proportional to the cell content, independently of the cell index. Sub-

stituting Eqs (6) through (8) into the equation of change (5), the system of differential equations

$$\begin{aligned} \dot{x}_0 &= \sigma x - \alpha x_0 \\ \dot{x}_i &= (\alpha + \beta(i-1))x_{i-1} - (\alpha + \beta i + \gamma)x_i \quad (i \in \mathbf{N}) \end{aligned} \tag{9}$$

is obtained, which the subsequent analysis is based on.

First, by summing Eqs (9) over the index $i \in \mathbf{N}_0$ an equation of change for the global system can be gained:

$$\dot{x} = (\sigma - \gamma)x - \lim_{i \to \infty} (\alpha + \beta i + \gamma)x_i. \tag{10}$$

From Eqs (2)–(4) it follows that

$$\lim_{i \to \infty} x_i = \lim_{i \to \infty} i x_i = 0, \tag{11}$$

hence Eq. (10) reduces simply to

$$\dot{x} = (\sigma - \gamma)x. \tag{12}$$

The solution is obviously

$$x = x(0)e^{(\sigma - \gamma)t} \tag{13}$$

where $x(0)$ is the content of the system at $t = 0$. The notorious exponential law is thus reaffirmed whether in the form of exponential growth ($\sigma > \gamma$), exponential decay ($\sigma < \gamma$), or, as a special case, of stationary state ($\sigma = \gamma$).

A question of substantially more interest is finding the distribution of the substance over the cells. To this end Eqs (9) should be transformed into ones containing the y_i-s instead of the x_i-s as dependent variables. Since

$$\dot{y}_i = \frac{\dot{x}_i x - x_i \dot{x}}{x^2} \quad (i \in \mathbf{N}_0), \tag{14}$$

using Eqs (9) and (12) one obtains

$$\begin{aligned} \dot{y}_0 &= \sigma - (\alpha + \sigma)y_0 \\ \dot{y}_i &= (\alpha + \beta(i-1))y_{i-1} - (\alpha + \beta i + \sigma)y_i \quad (i \in \mathbf{N}), \end{aligned} \tag{15}$$

a system of first order linear differential equations of extremely simple structure. The

solution can most straightforwardly be obtained by successive integrations resulting in

$$y_i = \bar{y}_i + \sum_{j=0}^{i} b_{ij} e^{-(\alpha+\beta i+\sigma)t} \quad (i \in N_0), \tag{16}$$

where the $\bar{y}_i$-s form the (unique) equilibrium solution of Eqs (15), obeying the recursion relations

$$\begin{aligned} \bar{y}_0 &= \frac{\sigma}{\alpha+\sigma} \\ \bar{y}_i &= \frac{\alpha+\beta(i-1)}{\alpha+\beta i+\sigma}\,\bar{y}_{i-1} \quad (i \in N); \end{aligned} \tag{17}$$

the coefficients b_{ij} are determined by the initial conditions. Since α and σ were claimed to be positive and β non-negative, all the exponential terms in Eq. (16) vanish as $t \to \infty$, and under arbitrary initial conditions the system evolves toward its asymptotically stable stationary distribution defined by Eqs (17). It is interesting to note that the distribution of the system is independent of the parameter γ (this is the consequence of the "uniform leakage" property). It is more important that while the distribution of the system tends to be time-invariant (stationary), the system as a whole assumes a stationary state only in the special case $\sigma = \gamma$, otherwise it grows or decays exponentially (cf. Eq. (13)).

Practically, starting from any given initial distribution, after several times the greatest characteristic time, $(\alpha + \sigma)^{-1}$, the system reaches a "steady-change" regime, in which the content of each cell grows or decays exponentially with identical characteristic time, $(\sigma - \gamma)^{-1}$, (in the special case of $\sigma = \gamma$ it remains constant), and the distribution among the cells is that of Eq. (17). Indeed, this distribution is the *Waring distribution*[1,22]

$$P(\xi = i) = \frac{a k^{[i]}}{(a+k)^{[i+1]}} \quad (i \in N_0) \tag{18}$$

with parameters

$$\begin{aligned} k &= \alpha/\beta, \\ a &= \sigma/\beta \end{aligned} \tag{19}$$

($k^{[i]}$ denotes the i-th ascending factorial product of k, i.e., $k^{[i]} = (k + i)!/k!$).

1.2. The Waring distribution

The expected value and the variance of the Waring distribution are

$$E(\xi) = \frac{k}{a-1}, \qquad \text{provided that } a > 1,$$
$$D^2(\xi) = \frac{ka(k+a-1)}{(a-1)^2(a-2)}, \qquad \text{provided that } a > 2, \tag{20}$$

respectively. Since the existence of a finite expected value was presupposed, the relation

$$\sigma > \beta \tag{21}$$

should necessarily hold.

In the special case of $\beta = 0$, the Waring distribution reduces to the geometric distribution

$$P(\xi = i) = q(1-q)^i \quad (i \in N_0) \tag{22}$$

with $q = \sigma/(\alpha + \sigma)$. This case refers to the absence of "cumulative advantage"

$$f_i = \alpha x_i \quad (i \in N_0); \tag{23}$$

this is the model of *Frank*[13] and *Coleman*[14].

In the case of $k = 0$ ($\alpha = 0, \beta \neq 0$), apparently, a degenerate distribution ($P(\xi = 0) = 1$, $P(\xi \neq 0) = 0$) is obtained. However, truncating the Waring distribution in zero and letting then k tend to zero results in

$$P(\xi = i \mid \xi > 0) = a\,B(a+1, i) \quad (i \in N) \tag{24}$$

($B(\cdot, \cdot)$ denotes the beta-function), i.e., the celebrated Yule distribution.[7,8,9,23] This case corresponds in our model to

$$f_i = \beta i x_i \quad (i \in N), \tag{25}$$

formalizing "Gibrat's law".[10]

A rather remarkable property of left-truncated Waring distributions is

$$P(\xi = n + i \mid \xi \geqslant n; k, a) = P(\xi = i; k + n, a) \quad (i, n \in N_0), \tag{26}$$

i.e., a Waring distribution with parameters k, a truncated from the left until n, is again a Waring distribution with parameters $k + n$, a. This is a very useful property in practical applications dealing with truncated samples.

Waring distribution obeys the frequency form of Zipf's law,[24] i.e.,

$$P(\xi = i) \to c i^{-(a+1)} \quad \text{as} \quad i \to \infty, \tag{27}$$

where c is a constant depending on the parameters of the distribution only and a is just

the parameter a of the Waring distribution. In other words, the Waring distribution asymptotically tends to a Lotka distribution[16] with exponent a+1. The significance of Zipfian distributions in scientometrics has been stressed recently by *Haitun.*[25]

Several methods of parameter estimation, including maximum-likelihood, are known from the literature.[3] A rather simple method is based on the relations

$$P(\xi = 0) = \frac{a}{a+k}$$
$$E(\xi) = \frac{k}{a-1} \tag{28}$$

Using the relative frequency, f_0, in place of the probability $P(\xi = 0)$ as well as the sample mean, $\bar{x}$, in place of the expected value $E(\xi)$, the estimators

$$\hat{k} = \frac{(1-f_0)\bar{x}}{f_0(\bar{x}+1)-1}$$
$$\hat{a} = \frac{f_0\bar{x}}{f_0(\bar{x}+1)-1} \tag{29}$$

of k and a, respectively, can be obtained. Although statistical properties of this estimation are not exactly known, its practical applicability has been well proved by our experience.

1.3. Repetitive events

A particular representation of the model outlined above is that of *repetitive events.* In this representation the system consists of elements which are classified into the "cells" according to the number of times they had undergone a certain event that may occur repeatedly. Elements with no occurence of the event belong to the 0-th cell, those with a single occurence to the 1-st, and so on. The "rules of game" are now the following:

(1) new entrants (with no occurence of the event) may enter the system at a rate proportional to the actual total number of elements in the system;

(2) the chance for the occurence of the event grows linearly with the number of events already had occurred;

(3) elements have equal chances to drop out from the system independently of the number of prior occurences of the event.

The system so defined behaves exactly as described above: the total number of elements varies exponentially in time; the distribution of the elements by the number of occurences of the event tends to a Waring distribution.

2. Applications

2.1. Scientific publication as repetitive event

Let the population considered be a community of scientists. Beyond doubt, the publication of a paper is an event that may occur repeatedly in a scientist's career. The question is, whether the "rules of game" postulated in our model could reasonably be considered to be met.

A source term proportional to the actual extent of the population, thus leading to an exponential growth is so commonly used in the literature that any specific reference seems to be superfluous. (It is worth noting that the model can readily be converted to obey the Gompertz, the Verhulst-Pearl or any other specific population growth law.)

"Cumulative advantage" property is also an often claimed feature of the scientific publication process. Its various verbal and formal expressions can be found in the literature as urn model with "contagion" of "cumulative advantage processes",[9] "Matthew effect"[26] "Gibrat's law"[10], or "success-breads-success phenomena".[27,6]

The "uniform leakage" property appears to be the most unprecedented among the three postulates. Indeed, its validity seems to depend heavily on the nature of the community under study. The assertion that scientists with different number of publications have equal chances to drop out from the community is trivially valid in the case of cumulative rolls (e.g. graduates of a given school), namely, in this case $\gamma = 0$: if a name has once got recorded on the roll, it cannot be deleted anymore. Though in some cases a greater number of publications renders dropout more in others less probable, in a great many of cases assumption of uniformity can be regarded as a suitable approximation.

2.2. Data sources and methods of analysis

Although innumerable samples of frequency distribution of publication productivity are commonly known beginning with those in the pioneering paper of *Lotka,*[16] all of them refer to some fixed time interval, and none of them follows the developmental dynamics of such distributions. We were thus forced to compile novel samples for our purposes. Two bibliographies, rather different both in volume and subject, were chosen to be analyzed: one on discrete distributions[22] (sample 1), the other one on frequency distributions of scientific performance[17] (sample 2). Annual publication productivity of each author was first recorded and then it was cumulated to yield eventually the frequency distribution of publication productivity up to each year.

Coauthored papers were counted differently in the two samples: in case of sample 1, full authorship has been assigned to each coauthor, whereas in case of sample 2, papers has been credited to the first author only.

Frequency distributions of publication productivity are truncated in zero by their very nature: frequency of scientists *not* publishing any paper up to a given year is practically undefined. Because of the property expressed in Eq. (26), there is no difficulty in fitting the Waring distribution to such samples either. Particularly, Eq. (29) of parameter estimation can easily be modified into

$$\hat{k}' = \frac{(1 - f_1')(\bar{x}' - 1)}{f_1'\bar{x}' - 1}$$
$$\hat{a}' = \frac{f_1'(\bar{x}' - 1)}{f_1'\bar{x}' - 1} \qquad (30)$$

where $k' = k + 1$ and $a' = a$ are the parameters of the truncated distribution (cf. Eq. (26)), f_1' and $\bar{x}'$ are the first relative frequency and the mean of the truncated sample, respectively.

Kolmogorov–Smirnov statistics (as proposed by *Coile*[28]) was used as an approximate test for goodness of fit (one should keep in mind that Kolmogorov–Smirnov theorem is rigorously valid for continuous distributions only).

2.3. Results

The time span of the samples analyzed has been twelve years (1955 through 1966 in case of sample 1, 1966 through 1977 in case of sample 2).

Number of authors plotted against the number of publications (cumulated) on a double logarithmic scale is presented in Figs 2 and 3.

The first few values of observed and calculated (based on Waring distribution) absolute frequencies together with the Kolmogorov – Smirnov statistics and the estimated parameters of the distributions are shown in Tables 1 and 2.

Growth of the total population of the authors as well as the number of authors publishing 1, 2, . . . papers up to the given year has been plotted on a semilogarithmic scale in Figs 4 and 5. Calculated regression lines are also indicated.

2.4. Discussion

Both series of samples strongly support the validity of the Waring model. The Kolmogorov–Smirnov statistics in Tables 1 and 2 all stay far below the value 1.36, which is the critical value at 0.05 significance level for accepting the hypothesis that the observed and calculated distributions are identical.

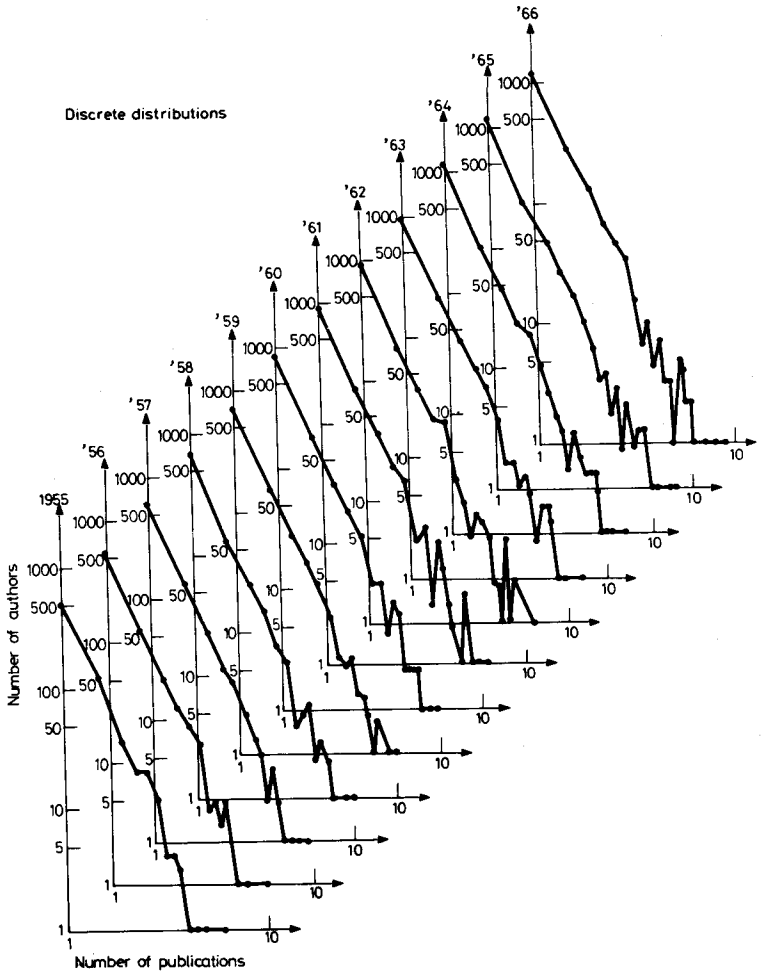


Fig. 2. Distribution of authors by publication productivity. Empirical data.
Sample 1 – "Discrete distributions"[22]

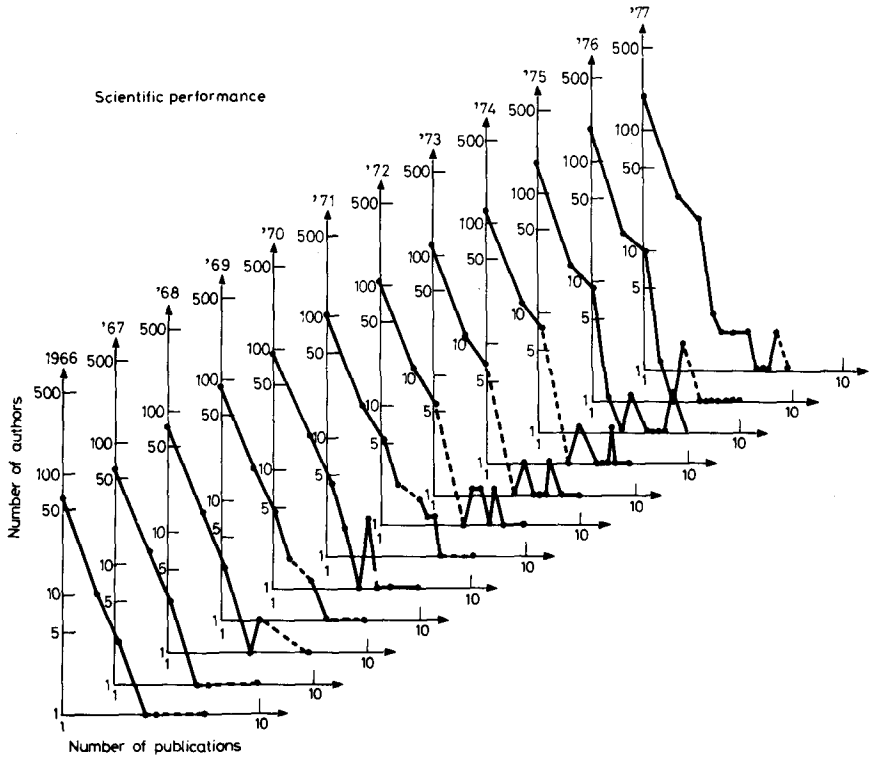


Fig. 3. Distribution of authors by publication productivity. Empirical data.
Sample 2 – "Scientific performance"[17]

Considering the parameters of the Waring distribution, one may observe that the k′ parameters of both series fluctuate around an average value, while there is a slight tendency in the time change of parameter a. In the case of sample 1, the value of a is constantly decreasing, in case of sample 2, it goes through a maximum. A parallel behaviour can be noticed in the growth rates: the growth curves in Fig. 4 are flattening, while those in Fig. 5 exhibit a clear inflection. This is just the kind of connection between growth rate and the parameter a predicted by Eq. (19).

However Figs 4 and 5 also testify that, in spite of the slight tendencies mentioned, growth can fairly well be represented by exponential functions (straight lines on the semilogarithmic plot). Even more, the growth coefficients for the number of authors of 1, 2, . . . papers are approximately equal to the time coefficient of the total growth, as stated by the model. The growth rate can thus be characterized by a unique time coefficient in each series of samples.

Table 1
Distribution of authors by publication productivity.
Observed and calculated values
Sample 1 – "discrete distributions"[22]

Number of publications	Number of authors							
	1955		1956		1957		1958	
	Observed	Calculated	Observed	Calculated	Observed	Calculated	Observed	Calculated
1	538	538.0	595	595.0	666	666.0	712	712.0
2	116	123.3	125	138.2	139	151.3	139	160.0
3	39	45.3	50	51.5	56	56.0	63	59.7
4	21	21.0	27	24.2	27	26.2	37	28.3
5	21	11.2	19	13.1	20	14.2	18	15.4
⩾ 6	26	22.2	33	27.0	35	29.4	40	33.6
Total number of authors	761		849		943		1009	
Total number of publications	1281		1464		1614		1759	
K–S statistics	0.497		0.505		0.397		0.664	
$\hat{k}'$	1.054		1.039		1.001		0.951	
$\hat{a}$	2.542		2.435		2.407		2.279	
	1959		1960		1961		1962	
	Observed	Calculated	Observed	Calculated	Observed	Calculated	Observed	Calculated
1	767	767.0	826	826.0	888	888.0	921	421.0
2	156	177.2	172	193.7	190	212.0	199	277.9
3	68	66.9	74	73.9	82	81.7	88	89.8
4	39	31.9	43	35.6	44	39.7	50	44.2
5	25	17.5	27	19.7	33	22.1	45	24.9
⩾ 6	44	28.4	51	44.2	57	50.6	64	59.2
Total number of authors	1099		1193		1294		1367	
Total number of publications	1949		2158		2386		2622	
K–S statistics	0.639		0.628		0.612		0.830	
$\hat{k}'$	0.983		0.986		0.998		1.025	
$\hat{a}$	2.271		2.219		2.182		2.116	

Table 1 cont.

Number of publications	Number of authors							
	1963		1964		1965		1966	
	Observed	Calculated	Observed	Calculated	Observed	Calculated	Observed	Calculated
1	996	996.0	1065	1065.0	1160	1160.0	1222	1222.0
2	218	245.1	234	261.3	249	280.3	256	294.5
3	92	96.4	102	103.0	114	110.5	123	116.4
4	58	47.4	53	50.8	61	54.7	64	57.8
5	39	26.7	42	28.7	40	31.1	47	33.0
≥ 6	72	63.4	82	69.2	89	76.4	94	82.3
Total number of authors	1475		1578		1713		1806	
Total number of publications	2822		3044		3333		3551	
K–S statistics	0.823		0.710		0.756		0.906	
$\hat{k}'$	1.017		1.000		0.961		0.946	
$\hat{a}$	2.114		2.077		2.017		1.980	

Table 2
Distribution of authors by publication productivity.
Observed and calculated values
Sample 2 – "Scientific performance"[17]

Number of publications	Number of authors							
	1966		1967		1968		1969	
	Observed	Calculated	Observed	Calculated	Observed	Calculated	Observed	Calculated
1	67	67.0	76	76.0	80	80.0	86	86.0
2	10	9.7	11	11.0	15	13.9	18	18.6
3	4	3.2	5	3.6	5	4.7	8	6.7
$\geqslant 4$	3	4.0	3	4.3	4	5.4	7	7.8
Total number of authors	84		95		104		119	
Total number of publications	126		140		158		194	
K–S statistics	0.120		0.144		0.137		0.083	
$\hat{k}'$	0.515		0.529		0.711		0.981	
$\hat{a}$	2.030		2.118		2.368		2.556	
	1970		1971		1972		1973	
	Observed	Calculated	Observed	Calculated	Observed	Calculated	Observed	Calculated
1	92	92.0	101	101.0	108	108.0	119	119.0
2	19	20.4	19	20.3	20	21.2	22	22.9
3	8	7.6	9	7.5	10	7.8	11	8.4
$\geqslant 4$	11	9.9	10	10.2	10	11.0	10	11.8
Total number of authors	130		139		148		162	
Total number of publications	227		243		261		284	
K–S statistics	0.228		0.373		0.321		0.275	
$\hat{k}'$	0.925		0.757		0.719		0.695	
$\hat{a}$	2.240		2.011		1.942		1.922	

Table 2 cont.

Number of publications	Number of authors							
	1974		1975		1976		1977	
	Observed	Calculated	Observed	Calculated	Observed	Calculated	Observed	Calculated
1	144	144.0	169	169.0	181	181.0	190	190.0
2	22	24.6	25	28.7	25	29.7	29	33.4
3	14	8.8	16	10.2	17	10.5	18	12.0
≥ 4	10	12.5	13	14.1	13	14.7	15	16.6
Total number of authors	190		222		236		252	
Total number of publications	324		372		394		426	
K–S statistics	0.254		0.564		0.306		0.283	
$\hat{k}'$	0.584		0.585		0.556		0.619	
$\hat{a}$	1.828		1.866		1.831		1.896	

Since, by the very nature of the samples, the leakage rate is zero ($\gamma = 0$), the time coefficients provide the values of the parameter σ directly (cf. Eq. (13)).

Using weighted least squares estimation the following values were obtained:

$\hat{\sigma} = 0.07485\ \text{year}^{-1}$ for sample 1
(doubling time = 13.36 year);

$\hat{\sigma} = 0.10094\ \text{year}^{-1}$ for sample 2
(doubling time = 9.91 year).

Disregarding the slight tendentious changes in parameter a, annual distribution parameters were averaged (using sample size as weight) and standard deviations were determined:

$\tilde{k}' = 0.9922 \pm 0.0326$
$\tilde{a} = 2.1778 \pm 0.1637$
for sample 1;

$\tilde{k}' = 0.6681 \pm 0.2158$
$\tilde{a} = 2.0006 \pm 0.1368$
for sample 2.

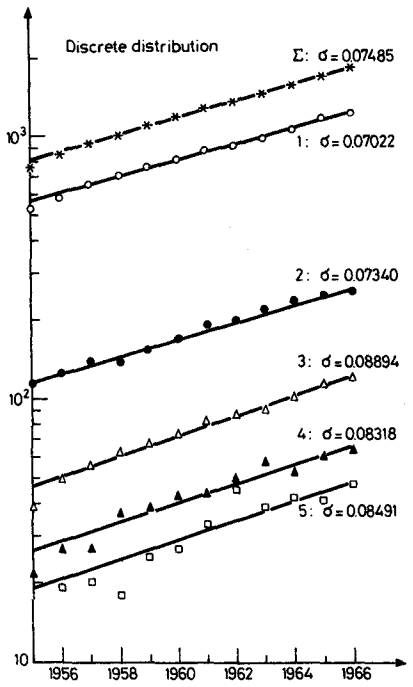


Fig. 4. Population growth of authors.
Sample 1 – "Discrete distributions"[22]

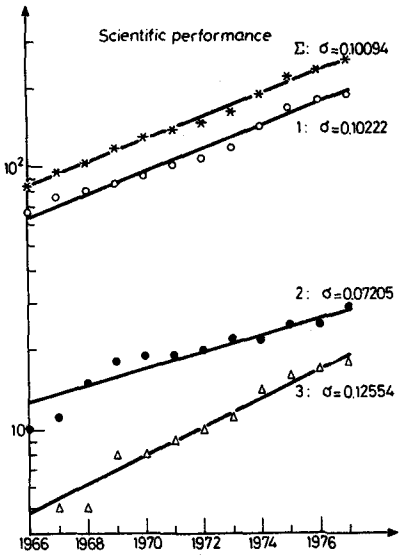


Fig. 5. Population growth of authors.
Sample 2 – "Scientific performance"[17]

Remarkably, in both cases a is very close to 2. According to Eq. (27), that means that they asymptotically tend to a generalized Lotka distribution with an exponent of value 3. Moreover, for sample 1 $k' \approx 1$ ($k \approx 0$) indicates an approximate Yule distribution (cf. Eq. (24)).

Eq. (19) enables us to calculate the "intrinsic" parameters, α and β, of the model. Parameter β is of particular interest, since, being the coefficient of i in the expression for f_i (Eq. (7)), it may rightly be considered the *cumulative advantage coefficient*. Practically, it might be regarded as a measure of enhancement affected by the previously published papers to later publication activity. Indirectly, it reflects the reward of publication in the given scientific community. The estimated values of β in the two samples are:

$$\hat{\beta} = \hat{\sigma}/\tilde{a} = 0.03468 \text{ year}^{-1} \text{ for sample 1,}$$
$$\hat{\beta} \quad = 0.05132 \text{ year}^{-1} \text{ for sample 2.}$$

These values suggest that in the field of research on scientific performance (between 1966–1977) publication has been more highly rewarded than in the field of discrete distributions research (between 1955–1966). To uncover the reason of this difference a deeper sociological study of the research fields in question is needed.

3. Conclusions

The main results of the paper may be summarized in the followings.

1. Population growth and distribution of authors by publication productivity are correlated features of a scientific community. A consistent dynamic model describing both phenomena has been established based on the theory of repetitive events.

2. Based on postulates of self-reproduction, cumulative advantage and uniform leakage, a model has been developed that tends to an asymptotically stable steady-change regime.

3. Under steady-change conditions the total population as well as the population of each stratum (the number of authors of 1, 2, . . . papers) has been found to grow exponentially with identical time coefficients.

4. Under steady-change conditions the distribution of authors by publication productivity has been proved to be a Waring distribution.

5. Cumulative advantage coefficient characterizing the reward of publication in a given scientific community has been derived and calculated.

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References

[1] J. O. IRWIN, The Place of Mathematics in Medical and Biological Statistics, *J. Roy. Stat. Soc.*, A 126 (1963) 1–45.

[2] J. O. IRWIN, The Generalized Waring Distribution Applied to Accident Theory, *J. Roy. Stat. Soc.*, A 131 (1968) 205–225.

[3] J. O. IRWIN, The Generalized Waring Distribution, Part I. *J. Roy. Stat. Soc.*, A 138 (1975) 18–31, Part II. *J. Roy. Stat. Soc.*, A 138 (1975) 204–227, Part III. *J. Roy. Stat. Soc.*, A 138 (1975) 374–384.

[4] G. HERDAN, *Quantitative Linguistics*, Butterworths, London, 1964.

[5] R. E. WYLLYS, Empirical and Theoretical Bases of Zipf's Law, *Library Trends*, 30 (1) (1981) 53–64.

[6] J. TAGUE, The Success-Breeds-Success Phenomenon and Bibliometric Processes, *J. Am. Soc. Inf. Sci.*, 32 (4) (1981) 280–286.

[7] H. A. SIMON, On a Class of Skew Distribution Functions, *Biometrika*, 42 (1955) 425–440.

[8] M. G. KENDALL, Natural Law in the Social Sciences, *J. Roy. Stat. Soc.*, A 124 (1961) 1–16.

[9] D. de SOLLA PRICE, A General Theory of Bibliometric and Other Cumulative Advantage Processes, *J. Am. Soc. Inf. Sci.*, 27 (5) (1976) 292–306.

[10] Y. IJIRI, H. A. SIMON, Skew Distributions and the Sizes of Business Firms, North-Holland Publ. Co., Amsterdam, 1977.

[11] D. O. O'CONNOR, H. VOOS, Empirical Laws, Theory Construction and Bibliometrics, *Library Trends*, 30 (1) (1981) 9–20.

[12] J. J. HUBERT, General Bibliometric Models, *Library Trends*, 30 (1) (1981) 65–81.

[13] R. FRANK, Brand Choice as a Probability Process, *J. Business*, 35 (1) (1962).

[14] J. S. COLEMAN, *Introduction to Mathematical Sociology*, Collier-Macmillan Ltd., London, 1964.

[15] E. XEKALAKI, Chance Mechanisms for the Univariate Generalized Waring Distribution and Related Characterizations, In: C. TAILLIE, G. P. PATIL (Eds), *Statistical Distributions in Scientific Work*, Vol. 4 – *Models, Structures, and Characterizations*, D. Reidel Publ. Co., Dordrecht, 1981. pp. 157–171.

[16] A. J. LOTKA, The Frequency Distribution of Scientific Productivity, *J. Washington Acad. Sci.*, 16 (12) (1926) 317–323.

[17] J. VLACHY, Frequency Distributions of Scientific Performance. A Bibliography of Lotka's Law and Related Phenomena, *Scientometrics*, 1 (1) (1978) 109–130.

[18] R. HJERPPE, A Bibliography of Bibliometrics and Citation Indexing & Analysis, Report TRITA–LIB–2013, Stockholm, 1980.

[19] R. HJERPPE, Supplement to a "Bibliography of Bibliometrics and Citation Indexing & Analysis" (TRITA–LIB–2013), *Scientometrics*, 4 (3) (1982) 241–281.

[20] A. PRITCHARD, G. R. WITTIG, *Bibliometrics. A Bibliography and Index.* Vol. 1: 1874–1959, ALLM Books, Watford, 1981.

[21] W. G. POTTER, Lotka's Law Revisited, *Library Trends*, 30 (1) (1981) 21–39.

[22] G. P. PATIL, S. W. JOSHI, *A Dictionary and Bibliography of Discrete Distributions*, Oliver & Boyd Ltd., Edinburgh, 1968.

[23] A. I. YABLONSKY, On Fundamental Regularities of the Distribution of Scientific Productivity, *Scientometrics*, 2 (1) (1980) 3–34.

[24] M. WOODROOFE, B. HILL, On Zipf's Law, *J. Appl. Prob.*, 12 (1975) 425–434.

[25] S. D. HAITUN, Stationary Scientometric Distributions, Part I. Different Approximations, *Scientometrics*, 4 (1) (1982) 5–25, Part II. Non-Gaussian Nature of Scientific Activities, *Scientometrics*, 4 (2) (1982) 89–104, Part III. The Role of the Zipf Distribution, *Scientometrics*, 4 (3) (1982) 181–194.

[26] R. K. MERTON, The Matthew Effect in Science, In: N. W. STORER Ed., *The Sociology of Science*, Univ. Chicago Press, Chicago, 1973. pp. 439–459.
[27] I. K. R. RAO, The Distribution of Scientific Productivity and Social Change, *J. Am. Soc. Inf. Sci.*, 31 (2) (1980) 111–121.
[28] R. C. COILE, Lotka's Frequency Distribution of Scientific Productivity, *J. Am. Soc. Inf. Sci.*, 28 (6) (1977) 366–370.